Prediction of Winning Probability and Ranking Generation in Horse Racing Using Machine Learning on Pre-Race Data

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***Abstract*—Preliminary abstract. This paper states the problem, the motivation and the proposed approach; it reports no experimental results, which are deferred to a later stage of the project.**

Horse race prediction has traditionally been framed either as an independent binary classification problem over individual runners or as a market-efficiency question in which tote odds serve as the main predictor. Neither framing exploits the structure of the task: a race is a closed field in which a known set of competitors is placed in a complete order. We propose a learning-to-rank formulation over the full finishing order, trained exclusively on information published before the starting gate opens, using the Hong Kong Jockey Club records from June 1997 to August 2005. Admissibility is decided by an explicit column-by-column leakage audit that removes post-race attributes such as finishing times, sectional positions and payouts, and the evaluation uses a chronological split in which every race remains an indivisible unit. Historical form is built from strictly earlier races. Ranking scores are converted to win probabilities by a calibration step fitted on the validation split, so that discrimination and calibration can both be reported. Model quality is assessed with ranking metrics together with race-level top-1 accuracy, mean reciprocal rank, log loss and Brier score, and the whole ladder is compared against the historical closing odds as an external market benchmark.

***Palabras clave*—machine learning, sports prediction, learning to rank, probability calibration, data leakage**

I. Introduction

Horse racing occupies a distinctive position within sports analytics, unlike most team sports, each race is a closed competition among a small set of participants with a single unambiguous outcome recorded down to the finishing order of every runner. This structure has made horse racing a recurring test bed for outcome prediction research since at least the 1980s [1] and it remains an active area of application for machine learning today. Driven largely by the scale of the betting industry built around it, the Hong Kong Jockey Club alone organizes close to 700 races per season across its two tracks, generating betting turnover in the billions of dollars annually. This combination of a well defined prediction target, a mature and data rich domain, and clear practical stakes makes horse race outcome prediction a useful problem through which to examine how a modeling formulation not merely a choice of algorithm affects prediction quality.

As detailed more in Section II, the great majority of the machine learning literature on this problem formulates it as binary classification, each runner is labeled as winner or non winner and standard classifiers are trained to separate the two classes. This formulation discards information that is often already present in the data used: most race datasets, including the one used in this work, record the complete finishing order of every runner, not merely the identity of the winner. Reducing this ordinal signal to a single binary label is a modeling choice, not a constraint imposed by the data. It is also, we argue, a contributing factor to a pattern visible across several of the reviewed studies: reported classification metrics that are difficult to reconcile with the extreme class imbalance inherent to the problem, or with the small scale of the datasets used to obtain them.

This motivates the central question addressed by this work:

> *Does formulating horse race outcome prediction as a learning-to-rank problem, trained to exploit the complete finishing order of each race rather than a binary win/loss label, improve the identification of race winners relative to the binary classification formulation predominant in the literature?*

The working hypothesis is that a learning to rank model, will outperform binary classification baselines on top 1 race accuracy because it is trained on an objective which is correctly ordering all runners of a race, that is strictly more informative than a binary label and because it does not require the class rebalancing procedures that binary classification approaches rely on to compensate for the natural imbalance of a one winner per race outcome.

The specific objectives of this work are to:

1) Define a race-level prediction framework for horse-racing outcomes that represents the relative performance of competing horses within each race and establishes a relevance formulation grounded in established betting categories rather than arbitrary discretization.
2) Develop a historical feature-engineering procedure that constructs race-time predictors exclusively from information available before each race, while preserving

horses and other entities with limited historical observations and preventing temporal leakage.
3) Establish a structured baseline ladder to separately assess the contribution of race-level grouping and the additional information provided by the complete finishing order, allowing the effect of each modeling decision to be evaluated independently.
4) Evaluate the proposed approach against the established baseline ladder and relevant results from the literature using reproducible experimental procedures and evaluation metrics appropriate to the prediction task.

The remainder of this document is organized as follows. Section II reviews prior work along three axes: approaches to the horse racing prediction problem in general, the technical methods relevant to machine learning, and studies using the dataset or comparable datasets employed in this work. The subsequent section III describe the literature baseline and the procedures used to support a reproducible evaluation. Section IV presents the proposed methodology, beginning with the formal definition of the prediction task and continuing with the relevance formulation, feature construction, model selection, experimental design, baseline ladder, risks, and evaluation metrics. Section V provides the technical analysis of the dataset, including its data dictionary, leakage audit, data-quality assessment, outlier detection, class imbalance, and feature distributions and correlations. Section VI discusses the ethical considerations associated with the potential use and broader context of the proposed system. Finally, Section VI-G discloses the use of generative AI tools throughout the development of the work and describes the verification procedures applied by the authors.

II. Related Projects

A. Axis A: The Problem Domain

Predicting the outcome of a horse race is framed in the literature in two largely disconnected ways: as a supervised classification problem over pre-race tabular features, and, less frequently, as a market-efficiency question about whether bookmaker odds already encode most of the predictable signal. These objectives are not interchangeable: they involve different target variables, evaluation procedures, and interpretations of predictive performance. Our review found supervised learning over pre-race tabular information is a recurring approach, while the formulation of the race as a direct within-race ranking problem is less explicitly addressed.

Classical statistical models

Early and still-recurring approaches rely on regression rather than modern ML. Li et al. [2] fit linear and log-linear regression models on going (track condition), distance and class to predict winning times at three UK racecourses, reporting $R^2 \approx 0.995$ for the fitted times themselves. However, the objective of the study is to audit whether officially reported track conditions are consistent with observed performance, not to predict the winner or establish a ranking of the horses; they explicitly note this is a starting point and suggest tree-based ML as a natural extension. This distinction isimportant because it illustrates a recurring pattern: regression-based work in this domain has mostly targeted auxiliary quantities (going, finishing time) rather than the win/place outcome itself, which is the kind of outcome-prediction aimed to solved in this project.

Machine learning approaches

The reviewed horse-racing studies show a range of supervised learning approaches and experimental settings. Davoodi and Khanteymoori [3] use artificial neural networks to predict the finishing time of individual horses at Aqueduct Race Track in New York. Their dataset contains 100 races from January 2010, and the authors compare five supervised training algorithms. They report an average accuracy of approximately 77%. The study therefore provides evidence that nonlinear supervised models can be applied to horse-racing data, but its small temporal and geographic scope limits conclusions about generalization.

Schumaker [4] studies harness racing at Northfield Park in Ohio using Support Vector Regression and historical race-performance variables. A key contribution of the study is its explicit temporal separation: data from October 2009 through September 2010 are used for training, while October through December 2010 are reserved for testing. The study also compares different race history windows and reports that the Time 4–3 configuration performs substantially better than the Time 7–3 configuration for the harness-racing setting. In particular, Time 7–3 achieves 35.42% accuracy, whereas Time 4–3 achieves 50.44% across the reported wagering analysis. These results shows that feature-history configurations developed for one racing subtype should not be assumed to transfer directly to another without validation. An important clarifition is that with the term Time 4–3, for example, we refer to the previous races being considered as reference for the study settings, in this case 4–3 considers from 4th to 3th previous race.

Gupta and Singh [5] examine horse-race prediction using data collected from a horse-racing website in India. Their dataset contains 14,750 records and 56 attributes, with wins representing only 11.8% of the observations. The authors report that a Random Forest model trained without imbalance handling can achieve high accuracy while failing to identify the minority class, whereas applying SMOTE improves the reported classification results to 93.1% accuracy, 92.9% F1-score (balance between precision and recall), and 97.6% ROC-AUC (ability to distinguish between classes across different decision thresholds). This study highlights an important domain characteristic: accuracy alone can be misleading when the target distribution is highly imbalanced, making class-sensitive metrics and appropriate training procedures important for horse-racing prediction. A related study by Gupta and Singh [6] compares K-nearest neighbors (KNN), logistic regression, Random Forest, Gaussian Naive Bayes, AdaBoost, and Bagging, together with Information Gain and Chi-Square feature- selection procedures. The authors report that the effectiveness of feature selection depends on the selection strategy, although, the Random Forest provides the best performance among the evaluated models across the despite feature-

selection settings, followed by AdaBoost as another good option.

Market efficiency and odds as a predictor

Colle [7] notes that horse-racing prediction research is comparatively scarce next to mainstream team sports precisely because commercial betting syndicates keep their methods private, and traces the field's statistical roots to pari-mutuel market modelling from the 1980s. Chavan et al. [8] summarize a study on Champ de Mars (Mauritius) in which betting odds alone reached only around 7.4% correct-rank prediction, which they interpret as evidence that odds are not, by themselves, a strong predictor of finishing order on that track.

What is comparatively settled

Across the studies we read, tree-ensemble and kernel methods consistently outperform plain linear/logistic baselines on tabular horse-racing features once class imbalance is addressed, and feature sets built from pre-race information (weight, draw, jockey/trainer history, ratings) are treated as the standard input space; nobody in our corpus argues that these features are irrelevant. What remains open is precisely what our project targets: first, every horse-racing-specific study we found evaluates on a single track, a short time window (weeks to a few years) or a single jurisdiction, so generalization across time and venues is essentially untested; second, none of the studies we read formulate the problem as a within-race ranking task (predicting an order over the runners of a given race) rather than an independent per-horse binary classification, even though a race is intrinsically a comparison among a fixed, known group of competitors; and third, explicit prevention of temporal data leakage is rare. Only Schumaker [4] uses a clean chronological split, while the others do not describe how (or whether) they avoided mixing future information into training folds. These three gaps are picked up again, together with the dataset-specific ones, in Section II-E.

Synthesis of the domain

The reviewed literature does not support the claim that a single modeling approach has been established as universally superior for horse-racing prediction. Instead, the studies demonstrate that the problem has been formulated in different ways, including continuous prediction of performance variables, individual-horse outcome prediction, and wagering-oriented prediction. Tree-based ensembles, neural networks, support-vector methods, and linear models all appear in the reviewed literature, but their reported results depend strongly on the target variable, dataset, racing subtype, feature construction, and evaluation protocol.

Two methodological patterns are particularly relevant to the present project. First, several studies operate within a single racecourse, jurisdiction, or relatively short observation period, which limits the evidence available about generalization across time and venues. Second, the reviewed studies use different target definitions and evaluation procedures, making direct comparisons of reported accuracy difficult. Schumaker [4] provides a useful counterexample by explicitly separating historical training data from a later testing period, illustrating the importance of evaluating a model under a temporal setting when the objective is future prediction.

These observations motivate two research gaps for the present project. First, the generalization of horse-racing prediction models across a longer historical period and a different data setting remains insufficiently characterized in the reviewed literature. Second, although race outcomes are inherently comparative because several horses compete in the same event, the reviewed studies primarily evaluate individual-horse predictions or wagering decisions rather than explicitly formulating the task as a within-race ranking problem. These gaps provide the domain-level motivation for the following axes. The techniques axis evaluates which classical machine-learning methods are appropriate for structured racing data and under what assumptions, while the dataset axis examines prior work using the selected Hong Kong racing data and establishes the most appropriate baseline. Together, these analyses will determine which aspects of the problem can be addressed by the proposed study and which remain outside its scope.

B. Axis B: Techniques

Gradient Boosting and Random Forests versus neural networks on tabular data

Tree-based ensembles are strong candidates for the present project because they can capture nonlinear relationships and interactions between variables without requiring a predefined functional form. The literature also provides evidence that their performance on structured tabular data can be competitive with, and in some settings superior to, more complex models. Hall and Rasheed [9] conduct a systematic review of 79 studies comparing tree-based machine learning (TBML), including Random Forest, XGBoost, LightGBM, and CatBoost, with deep learning across multiple domains. TBML methods are the most frequent top performers in their review and require substantially less training time on average, with reported differences of two to four orders of magnitude. However, their evidence comes primarily from time-series forecasting rather than horse-racing prediction, and the authors emphasize that performance depends on factors such as dataset characteristics, temporal structure, and experimental design. Therefore, the review supports considering tree-based methods as strong baseline candidates, but does not establish their superiority for the present dataset.

Evidence from other structured-data domains further supports this choice while illustrating the importance of empirical comparison. Dumitrescu et al. [10] compare Random Forest, logistic regression, SVM, and neural networks across four credit-scoring datasets. On the Kaggle *Give Me Some Credit* dataset, for example, Random Forest and their tree-augmented logistic model PLTR achieve AUC values of 0.853 and 0.857, respectively, compared with 0.752 for the neural network. These results show that a more complex model does not necessarily provide better predictive performance on structured data. Similarly, Yuan et al. [11] review machine-learning approaches for sports-injury prediction and report strong results from gradient-boosted tree methods while also

identifying settings in which simpler logistic-regression models perform comparably or better. Together, these studies support treating model complexity as an experimental variable rather than assuming that neural networks will provide the best performance. Within racing-related applications, Kholkine et al. [12] use XGBoost to predict road-cycling race performance. The authors motivate the use of XGBoost by its predictive performance, computational efficiency, and ability to handle missing values. Their application is particularly relevant to the present study because missing values can represent meaningful events, such as riders who did not finish a previous race. This illustrates why missing historical information should be analyzed according to its meaning before an imputation strategy is selected.

Ranking within a group, not independent binary classification

A horse race produces a total order over a small, closed set of competitors, which is a ranking problem, not a set of independent win/lose labels, yet none of the horse-racing-specific studies we read in II-A model it that way; all of them predict a per-horse label or score and then sort. This distinction is important when defining the prediction target: a model that estimates the probability of winning for each horse is different from a model whose objective is to predict the relative order of all horses within a race.

The distinction between pointwise prediction and group-level ranking is illustrated by work outside and within horse racing. Peng et al. [13] formulate rank-position forecasting in IndyCar as the prediction of future rank positions within a race. Their baselines include pointwise machine-learning regression models such as Random Forest, SVM, and XGBoost, as well as a naive baseline called CurRank, which assumes that the current rank will not change. Their proposed RankNet explicitly models rank-position sequences and reports improvements over the baselines in the evaluated forecasting tasks. In particular, the authors report that RankNet-MLP exceeds the CurRank baseline by 7% in Top-1 accuracy and 19% in MAE for the main forecasting task. These results do not demonstrate that ranking models are necessarily superior for horse racing, but they show that the formulation of the target and the structure of the competitors can materially affect the prediction problem.

Within horse racing, Silverman and Suchard [5] provide a complementary approach through regularized conditional logistic regression with frailty, which incorporates dependence among competitors within the same race when predicting the winner. Although their model does not predict the complete finishing order, it demonstrates that race-level structure can be incorporated directly into the statistical formulation rather than treating each horse as an entirely independent observation.

These studies suggest that the race should be considered the natural grouping unit when defining the prediction task and evaluating the resulting model. However, they do not by themselves establish which formulation is most appropriate for the present dataset. A binary classification formulation is suitable if the objective is to estimate whether an individual horse wins its race. If the objective is to produce an ordering of the competitors, a ranking-oriented formulation would be more directly aligned with the target.

Calibration and explainability

If the final system produces probabilities rather than only a predicted ordering, discrimination metrics alone are insufficient. Vuko et al. [14] explicitly evaluate calibration in addition to discrimination and argue that model selection should consider whether predicted probabilities correspond adequately to observed frequencies. Accordingly, log loss, Brier score, or an appropriate calibration analysis can complement ranking and classification metrics in the experimental protocol.

For interpretability, Kholkine et al. [12] use SHAP-based feature attribution to inspect the contribution of predictors in their XGBoost model. SHAP should be regarded as an interpretability tool rather than as evidence that XGBoost is inherently superior. Its role in project is to analyze which pre-race features contribute to model predictions after a model has been validated, while avoiding causal interpretations from feature-attribution values alone.

Class imbalance and evaluation

Predicting the winner of a race produces an intrinsically imbalanced target when each horse is represented as an individual observation. Gupta and Singh [5] illustrate this problem clearly: only 11.8% of their observations correspond to winners. After applying SMOTE, tetthey report substantially higher F1 and ROC values for Random Forest than in the unbalanced setting. However, synthetic oversampling must be treated carefully when observations belong to races, because generating synthetichorses without respecting race membership can distort the competitive structure. Hence, imbalance-handling methods such as class weighting or resampling should be evaluated experimentally rather than adopted solely because they improved a result in another dataset.

Overall, the researches supports a candidate set centered on logistic regression and tree ensembles, with Random Forest and gradient boosting methods such as XGBoost as flexible nonlinear alternatives. KNN, Naive Bayes, AdaBoost, and Bagging can be used as secondary comparison models if computational and experimental scope allow. The final selection should be determined empirically using a common temporal protocol and metrics aligned with the final target definition. This approach avoids assuming that the most complex or most frequently reported algorithm is necessarily the best choice for the Hong Kong racing dataset.

For these reasons, the selected track is classical machine learning over structured tabular data. Consequently, the relevant question is not which algorithm is universally the best, but which model families provide an appropriate balance between predictive capacity, assumptions, computational cost, and interpretability for the structure of horse-racing data. The literature reviewed for this project suggests that tree ensembles, regularized linear models, and other classical classifiers are viable candidates, while model choice should remain dependent on the target definition, feature representation, and evaluation protocol.

C. Axis C: The Dataset

The dataset specification

We use the *Horse Racing in HK* dataset published on Kaggle by Graham Daley, distributed as two tables, `races.csv` and `runs.csv`. We verified the following directly from the files rather than relying on the dataset page description: 6,349 races and 79,447 individual horse runs, spanning 1997-06-02 to 2005-08-28. `races.csv` records per-race attributes (venue, distance, going, class, sectional and finishing times, dividends); `runs.csv` records per-horse-per- race attributes (draw, declared/actual weight, horse rating, gear, trainer and jockey identifiers, sectional positions, and finishing result). Feature availability is consistent with a pre-race prediction setting for most columns (draw, weight, rating, jockey/trainer identity are known before the race), while several columns (sectional positions, finishing time, dividends) are only known during or after the race and must be excluded from any predictive feature set.

Comparable work on Hong Kong racing

Silverman [5] performs a hierarchical Bayesian analysis of Hong Kong horse racing: 36,006 observations from 2,973 races, modelling each horse's running speed and using the posterior estimates to identify the horse expected to be fastest in each race, reaching 21.6% accuracy against the actual outcome. This is, to our knowledge, the closest jurisdiction-matched published result available to us, same racing authority (Hong Kong) as our dataset, though roughly half our race count (2,973 versus our 6,349) and with the exact period left unspecified in the source available to us, so we cannot confirm whether it overlaps with our 1997–2005 window or covers a different span of Hong Kong Jockey Club history. The target itself also differs subtly from ours: Silverman's model identifies the horse with the *highest estimated running speed*, which is correlated with but not identical to the *official race winner* (a horse can run fastest on average and still lose due to pace, positioning, or trip). We treat this difference as a documented protocol gap, not a disqualifying one, and use it as our primary literature baseline. The same author, with Suchard, later moved from this Bayesian speed-estimation framing toward a discriminative conditional-logit model with frailty terms, which we read as the same research line converging toward a ranking- style formulation closer to what we plan to adopt.

Analogous datasets, for secondary reference

Beyond the jurisdiction- matched Silverman result above, we also keep structurally similar horse-racing datasets from other jurisdictions as secondary analogs, both to triangulate what accuracy range is achievable with modern classical ML (Silverman's own result predates tree-ensemble methods) and because the assignment allows analogous datasets to complement. The closest analogs by scale are Gupta and Singh [5], [6] (Turf Club of India, 14,750 records, 2017–2019, 11.8% positive class) and Davoodi and Khanteymoori [3] (Aqueduct, NY, 100 races). They cover a different racing jurisdiction, wagering structure, and (in Davoodi and Khanteymoori's case) a much shorter time window, so any accuracy figure taken from them should be treated as a loose analog, not a like-for-like target; we discuss this comparability gap explicitly in the baseline.

Known issues

Three dataset-level limitations are worth stating plainly. First, temporal coverage ends in 2005, so any claim about current racing dynamics (field sizes, betting patterns, track conditions) should be qualified as historical. Second, the positive-class rate is structurally low (roughly one winner per race of typically 10–14 runners), mirroring the imbalance documented in Gupta and Singh [5] for a different jurisdiction, so imbalance handling is not optional. Third, several columns in `runs.csv` (such as gear, sectional positions for shorter races) contain structurally missing values which must be documented and imputed (or excluded) accordingly rather than dropped by default.

D. Tabla Comparativa

The table summarizes the main experimental studies reviewed across the three axes, including their modeling techniques, datasets or environments, primary evaluation metrics, reported results, and identified limitations. The comparison shows substantial variation in both prediction tasks and experimental protocols. Horse-racing studies generally use different jurisdictions, dataset sizes, target definitions, and evaluation procedures, while studies from adjacent domains provide evidence for alternative modeling techniques and problem formulations. Therefore, it serves primarily to organize the evidence used to identify the research gaps and methodological considerations discussed in the following sections, rather than to establish a single ranking of the reviewed models. to see more go to TABLE I

The comparison table uses several standard machine-learning abbreviation and terminology. *Artificial Neural Network* (ANN) refers to a neural-network model for learning nonlinear relationships between input features and a target. *Support Vector Regression* (SVR) is a support-vector method for predicting continuous outcomes, while *Support Vector Machine* (SVM) is its commonly used classification counterpart. *Random Forest* (RF) is an ensemble of decision trees whose predictions are aggregated, and *Gradient Boosting* (GB) sequentially builds trees to correct previous prediction errors. *XGBoost* is an optimized gradient-boosting implementation.

SMOTE (Synthetic Minority Over-sampling Technique) is a class-imbalance preprocessing method that generates synthetic minority-class examples. Other terms in the table describe methodological combinations rather than individual predictive models. For example, *SHAP* (SHapley Additive exPlanations) is an interpretability method used to estimate the contribution of individual features to model predictions. These terms describe either predictive models or supporting techniques and should not be interpreted as equivalent types of algorithms.

TABLE I: Comparison of Related Work

Ref.	Year	Technique	Dataset / environment	Primary metric	Result	Identified limitation
Axis A — domain (Int. 1)						
[3]	2010	ANN (5 training algorithms)	Horse racing, Aqueduct NY (100 races)	Accuracy	77.0 %	Single track; one month of data
[4]	2018	SVR	Harness racing, Northfield Park (Ohio)	Accuracy	50.4 %	Single harness-racing track; performance depends on race-history window
[5]	2023	R F + SMOTE	Horse racing, Turf Club of India (14,750 records)	Accuracy / F1-score / ROC-AUC	93.1 % / 92.9 % / 97.6 %	Single site; different jurisdiction from ours
[2]	2025	Linear / log-linear regression	Horse racing, 3 UK racecourses	R^2	0.995	Models winning-time consistency rather than the race winner
Axis B — techniques (Int. 2)						
[12]	2020	XGBoost + SHAP	Road cycling, Tour of Flanders	Top-10 hits / 10 races	6/10 (2018)	Small dataset; several performance across race editions
[13]	2020	RankNet (LSTM encoder-decoder) vs. RF/SVM/XGBoost	Car racing, IndyCar (Indy500)	Top-1 accuracy / MAE	79.0 % / 0.94	Classical ML baselines did not consistently beat the no-change baseline
[10]	2022	Penalised Logistic Tree Regression (PLTR)	Credit scoring (4 datasets)	AUC	85.7 %	Cross-domain; several model performance across datasets
[14]	2025	Logistic regression / Lasso / Ridge / Elastic Net	Going-concern prediction, Croatia (891 obs.)	AUC-ROC	84.2 %	Small, imbalanced dataset; differences between models not statistically significant
[15]	2026	SVM + RF + GB ensemble	Formula 1 race results (OpenF1, 1,635 records)	Macro F1-score	77.8 %	Winner class hardest; only 2023–2024 seasons
[?]	2013	Conditional logistic regression + frailty (GPU)	Horse racing, 3,681 races	GPU/OpenMP speedup	by a time factor of 2.68	No accuracy or AUC reported; computational result only
Axis C — dataset (Int. 3)						
[5]	2012	Hierarchical Bayesian model (running speed)	Horse racing, Hong Kong (2,973 races)	Accuracy (fastest horse)	21.6 %	Predicts fastest horse, not official winner; period unspecified
[8]	2024	Logistic Regression / Decision Tree / RF / SVM / Dummy Classifier	Horse racing, 6,348 races / 79,447 records	Accuracy / F1-score	92.0 % / 51.0 %	Random 80/20 split; Dummy Classifier also achieves 92.0 % accuracy

E. Identified Research Gaps

Gaps identified in the literature

First, horse-racing-specific research is generally restricted to a single racing venue, jurisdiction, or relatively short observation period. Davoodi and Khanteymoori [3] use one month of data from a single track, Schumaker [4] uses 15 months from one harness-racing track, and Gupta and Singh [5], [6] use three seasons from a single Indian racing operator. Consequently, the literature provides limited evidence about whether reported models generalize across venues or racing jurisdictions. Second, Schumaker [4] shows that a race-history feature window developed for greyhound racing performs poorly when transferred to harness racing, with wagering accuracy decreasing from 50.4% to 35.4%. This provides direct evidence that feature configurations should be validated for the specific racing context rather than transferred without re- evaluation. Third, Yuan et al. [11] and Hall and Rasheed [9] emphasize that comparisons across studies are difficult because differences in features, data splits, preprocessing, and evaluation procedures can obscure the effect of the modeling technique itself.

Patterns inferred by comparison

First, the horse-racing-specific studies considered in this project formulate prediction at the individual-horse level, using a binary or regression target and sorting the resulting predictions afterward. None of the reviewed horse-racing studies explicitly formulates the complete race outcome as a learning-to-rank problem. This observation is not presented as a limitation stated by the authors, but as an inference from the reviewed corpus; evidence from related racing and ranking domains discussed in the II-B, suggests that the grouping of competitors within a race may deserve explicit consideration. Second, only Schumaker [4] explicitly describes a chronological train/test split among the horse-racing studies reviewed. The remaining studies provide limited information about their temporal splitting procedures. Therefore, the extent to which their reported results would generalize to future races is difficult to assess. This observation is treated as a limitation of the available evidence rather than as proof that the reported experiments contain temporal leakage.

What this project addresses

The present project addresses these methodological gaps within the limits of the available Hong Kong dataset. We use a chronological train/validation/test protocol to evaluate performance on future races and restrict predictive features to information available before the race, reducing the risk of temporal leakage. We also investigate the race as the natural grouping unit and evaluate whether a ranking-oriented formu-

lation is appropriate for the prediction objective, rather than assuming that independent per-horse classification is sufficient. Feature configurations are selected and validated using the characteristics of the Hong Kong dataset rather than being transferred directly from another racing sub-type. Finally, calibration is considered alongside predictive performance to distinguish between ranking quality and the reliability of predicted probabilities

What remains out of scope

The present project addresses these methodological gaps within the limits of the available Hong Kong dataset. We use a chronological train/validation/test protocol to evaluate performance on future races and restrict predictive features to information available before the race, reducing the risk of temporal leakage. We also investigate the race as the natural grouping unit and evaluate whether a ranking-oriented formulation is appropriate for the prediction objective, rather than assuming that independent per-horse classification is sufficient. Feature configurations are selected and validated using the characteristics of the Hong Kong dataset rather than being transferred directly from another racing sub-type. Related studies on physiological and psychological stress [16], equestrian performance factors [17], and genetic breeding values [18] illustrate research directions that require information we do not have. Finally, calibration is considered alongside predictive performance to distinguish between ranking quality and the reliability of predicted probabilities.

III. Literature Baseline

Primary literature baseline

The best jurisdiction-matched published result we found is Silverman [5]: a hierarchical Bayesian model of running speed fit to 2,973 Hong Kong races (36,006 observations), reaching 21.6% accuracy in identifying the horse that actually turns out fastest. We use this result as the primary literature reference point for our evaluation, particularly for a per-race top-1 prediction task. However, the value should not be interpreted as a strictly like-for-like performance threshold because Silverman's prediction target is the horse with the highest estimated running speed rather than the official race winner recorded in our dataset. The 21.6% result therefore serves as a jurisdiction-matched reference point against which the difficulty of the prediction problem can be contextualized, rather than as a directly equivalent benchmark.

We document the following comparability gaps rather than treating the number as a like-for-like target, the main differences are as follow:

- Race count and period: 2,973 races compared with our 6,349 races. The exact calendar period of Silverman's dataset is not stated in the source available, so we cannot determine whether it overlaps with, precedes, or follows our 1997–2005 period observation window.
- Target definition: Silverman's model predicts the horse with the highest estimated running speed, whereas our target is the race winner recorded in our dataset; these outcomes are related but not identical, since the horse estimated to be fastest may not necessarily finish first in a particular race.
- Technique: Silverman uses a hierarchical Bayesian model for running speed rather than a discriminative classifier or a learning-to-rank model. Consequently, differences in model family, target definition, and evaluation protocol prevent a direct apples-to-apples comparison without reproducing both approaches on the same dataset.

We also note, without using it as a baseline, that Silverman and Suchard [5] later extended this line of work toward a discriminative conditional logit with frailty terms; it reports a computational speedup (GPU vs. OpenMP by a factor of 2.68) for that model but no accuracy or AUC figure, so it cannot be used quantitatively here (Section II-B).

Secondary literature baseline) Chavan et al. [8] provide the closest dataset-matched reference identified in our review. The authors explicitly state that they use the Hong Kong horse-racing dataset available on Kaggle, consisting of the `races.csv` and `runs.csv` files, with 6,348 races and 79,447 horse records. Their preprocessing merges the two files using the race identifier and encodes categorical variables such as venue and going before model training. The study evaluates a Dummy Classifier, Logistic Regression, Decision Tree, Random Forest, and Support Vector Machine using a random 80/20 train-test split. Random Forest achieves the highest reported precision (62%) and an accuracy of 92%, with 52% recall and a 51% F1-score. The authors also show that the Dummy Classifier achieves the same 92% accuracy by always predicting the majority *not winning* class, illustrating why accuracy alone is insufficient for this imbalanced prediction problem.

Secondary analogs

As additional reference points are provided by studies conducted in racing environments outside Hong Kong. Gupta and Singh [5] evaluate Random Forest with SMOTE on Turf Club of India data and report 93.1% accuracy and 97.6% ROC-AUC. Davoodi and Khanteymoori [3] evaluate five neural-network algorithms on 100 races from Aqueduct and report an average accuracy of 77%. Schumaker [4], working with harness racing at Northfield Park, reports 50.4% wagering accuracy using support vector regression. These results provide useful methodological context, but they are not directly comparable with the present study because they differ in jurisdiction, racing sub-type, dataset size, prediction target, and evaluation protocol. In particular, wagering accuracy and winner-classification accuracy should not be interpreted as equivalent measures. We therefore use these studies only as contextual evidence rather than as quantitative performance targets.

IV. Proposed Methodology

A. Problem Formalization

In most of the works reviewed in Sections II-A and II-B, horse race outcome prediction is formulated as a binary classification problem. This paper argues that this simplification discards

relevant information available in the data and contributes to the accuracy limitations observed in the literature. For this reason, it is proposed formulating the problem as a learning to rank problem or LTR, which is possible because the dataset used does not merely record a binary outcome for each runner, but their complete finishing position. This allows the model to leverage the full ordering of each race, rather than reducing the available signal to a single bit per runner.

Formally, let $r \in R$ be a race from the set of available races, and $H_r = \{h_1, h_2, \ldots, h_{n_r}\}$ the set of runners competing in r, where n_r is the number of runners in that race. Each runner $h_i \in H_r$ is described by a feature vector $\mathbf{x}_i \in R$ and has a relevance label y_i, derived from its final finishing position through:

$$y_i = g(\text{position}_i) \tag{1}$$

where g is a decreasing function that assigns higher relevance to better finishing positions. This formulation allows the model to leverage the complete ordering of each race, rather than only the identity of the winner.

The learning objective is then to find a scoring function f_θ, parameterized by θ, such that the ordering induced by $\{f_\theta(\mathbf{x}_i)\}_{i \in H_r}$ approximates the true finishing order of race r. An important detail that needs to be noted is that f_θ is evaluated independently for each runner, but is trained and validated by grouping observations by race, since the relevant comparison occurs exclusively among runners within the same race.

B. Model Selection

The choice of model was primarily guided by the need to directly optimize runner ordering, rather than a binary as it was mentioned before. Under this criterion, LambdaMART was selected which is one of the most widely used and robust algorithms in the field of learning to rank [19].

LambdaMART combines two components. The first is MART (Multiple Additive Regression Trees), an ensemble of gradient boosted decision trees (GBDT) trained sequentially, where each tree corrects the residual errors of the previous iterations [20]. The second component is the set of lambda gradients, which determine the direction and magnitude by which each runner's score should be adjusted; these are computed by multiplying the gradient of a pairwise loss by the actual change that the ranking metric would undergo if two runners swapped positions [20].

This approach is particularly well suited when the system is evaluated directly with ranking metrics and when a set of pre extracted and calculated features are already available, both conditions hold in this project.

The primary metric used to evaluate ranking quality is Normalized Discounted Cumulative Gain (NDCG), one of the most widely used listwise metrics for ranking systems, as it considers the complete list of results returned for each query[[19]]. In this case each race rather than evaluating each runner in isolation. For a race r, it is defined as:

$$\text{DCG@}k(r) = \sum_{i=1}^{k} \frac{2^{y_i} - 1}{\log_2(i+1)} \tag{2}$$

$$\text{NDCG@}k(r) = \frac{\text{DCG@}k(r)}{\text{IDCG@}k(r)} \tag{3}$$

where i indexes the position in the ranking predicted by the model (not the true finishing position), y_i is the true relevance of the runner placed at that predicted position, and IDCG@$k(r)$ is the maximum possible value of DCG@k for race r.

C. Projected Pipeline

1) **Data integration and parsing**: the dataset is provided as two related tables, one containing race-level attributes and the other containing runner-level attributes, these joined through a race identifier in a one-to-many relationship, since a single race involves multiple runners. Constructing a clear data dictionary at this stage is essential, as it determines which fields are usable downstream and prevents ambiguous or redundant fields from propagating.
2) **Historical features**: Since the source tables only have raw identifiers and no pre-aggregated from statistics historical performances must therefore be constructed.
   First, for any entity(horse, jockey, trainer) and race date t, only races strictly prior to t may be used in order to prevent temporal leakage. Second, in the database history is very unevenly ditributed. Horses have a median of 14 prior races, but around a 5.5% of observations are horse's debuts and around a 24% of horses have five races or fewer in total. Prior work has handled this by disrigarding horses with little history[[3]] but that throws away valuable data.
   Instead, for each entity e and statistic x, the proposed approach will use a shrinkage estimator that blends the entity's own recent history with the population average:

$$x_e(t) = \frac{n_e(t) \cdot \tilde{x}_e(t) + k \cdot \mu_x}{n_e(t) + k} \tag{4}$$

   where $n_e(t)$ is the number of prior races available for an entity e at a time t, $\tilde{x}_e(t)$ is a recency-weighted average of x over those prior races and k is a shrinkage strenght constant. Its important to note that recent races will weight more that old ones via a decay factor.
   With this formula both extremes will be handled without special-case rules since when an entity has a longer history $x_e(t)$ is driven mostly by $\tilde{x}_e(t)$ and when history is less pronounced it falls back smoothly to the population average.
3) **Relevance label construction**: the finishing position of each runner is mapped onto the graded relevance scale. Rather than adopting an arbitrary discretization, the thresholds are aligned with betting categories established

in the horse racing domain itself *win*, *place*, and *show* yielding:

$$y_i = \begin{cases} 3 & \text{if position}_i = 1 \\ 2 & \text{if position}_i \in \{2, 3\} \\ 1 & \text{if position}_i \in \{4, 5, 6\} \\ 0 & \text{otherwise} \end{cases} \tag{5}$$

This grading was verified against the empirical distribution of finishing positions in the dataset, which is approximately uniform across positions 1 through 12 ($\approx$8% of runners per position), confirming that no single relevance grade is trivially dominant and that the ranking task is non degenerate.

4) **Race grouping**: observations are sorted contiguously by race identifier, and a group-size vector is constructed because group misconfiguration fails silently rather than raising an explicit error, the total of the group sizes is validated against the total row count before every training run.
5) **Train and validation**: given that the dataset spans several racing seasons, races are partitioned chronologically rather than randomly, so that no validation or test race precedes, in time, any race used for training. A random split would place runners from the same underlying time series on both sides of the partition, reintroducing the leakage that the historical features were designed to avoid.
6) **Baseline ladder training**: models are trained in order of increasing complexity, from a non-learned baseline, through a flat binary classifier and a group-normalized (conditional logit) model, up to the proposed LambdaMART ranker. Training each intermediate model is not merely a formality; it isolates the specific contribution of respecting the group structure of a race from the contribution of exploiting the full finishing order, which would otherwise be conflated in a single end to end comparison.
7) **Evaluation**: model performance is assessed using the ranking metrics together with top-1 race accuracy, reported separately to allow direct comparison with the classification-based results reported in prior work.

D. Experimental protocol

Given that the data covers races between the dates of June 1997 and August 2005, a chronological partition is used to prevent information from future races from influencing predictions for past races via the historical variables. More specifically this work will use races from 11997 to 2003 for training, 2004 for validation and 2005 for testing. This partition preserves the integrity of each race as an indivisible unit which means that no runner from a given race is assigned to a different set than the other runners in that same race.

E. Baseline plan

To isolate and quantify the contribution of each design decision, the study will use the following hierarchy, ranging from lower to higher complexity.

- Market favorite, serving as a non machine learning reference.
- Flat binary classification(logistic regression, gradient boosting on wins and losses), representing the dominant approach in the reviewed literature.
- Conditional logit which isolates the effect of satisfying the constraint that race probabilities must sum to one.
- Literature baseline, corresponding to the test result reported on similar datasets studied in section [-eje A y B].
- Proposed model, LambdaMART.

F. Risks and Mitigation

TABLE II: Identified methodological risks and mitigation strategies.

Risk	Mitigation
Temporal leakage in historical features: computing a horse's, jockey's, or trainer's form using races that occur after the one being predicted.	All historical features are computed exclusively over races strictly prior to the target race, and data are split chronologically (train / validation / test) rather than randomly, so that no leakage can occur across the split.
Silent misconfiguration of race-wise grouping in the ranking model: an incorrectly built group vector does not raise an explicit error, yet invalidates training.	The sum of group sizes is programmatically validated against the total row count before every training run.
Informational leakage via market odds: win odds already reflect the aggregated judgment of informed bettors at race time, which may inflate apparent model performance if used uncritically.	Models are reported and compared both with and without odds-derived features, and this decision is discussed explicitly rather than treated as a neutral feature choice.
Absence of pedigree information (sire/dam lineage) in the dataset, despite its documented relevance in the domain literature.	Declared as an acknowledged limitation from the survey stage onward, rather than discovered later; not compensated for by proxy variables that could introduce unjustified assumptions.
Class imbalance in the win/non-win outcome ($\approx$8%).	Substantially mitigated by the learning to rank formulation itself, since the model never optimizes a binary objective over an imbalanced label; no synthetic resampling is required or applied.

V. Technical Analysis

This section reports the exploratory analysis performed before any model was trained. It is organised around one question: which information was genuinely available when the starting gate opened, and which was not. Feature admissibility, imputation and partitioning all follow from that question. Every figure quoted below was computed directly from the two source tables by a single scripted pass with a fixed random seed, so that the whole analysis can be re-executed end to

TABLE III: Condensed data dictionary

Fields	Meaning	Avail.
runs.csv, one row per runner		
race_id, horse_no, horse_id	Race key, saddlecloth number, horse key	PRE
horse_age, horse_country, horse_type	Age, country of origin, sex/type (gelding, mare, colt)	PRE
horse_rating, horse_gear	Official HKJC rating before the race; declared equipment	PRE
declared_weight, actual_weight, draw	Weight of horse plus equipment, assigned handicap weight, barrier	PRE
trainer_id, jockey_id	Pseudonymous identifiers of the connections	PRE
win_odds, place_odds	Final (starting-price) tote odds	MKT
result, won	Finishing position and win indicator (the target)	POST
lengths_behind, finish_time	Beaten margin and total running time	POST
position_sec1–6, behind_sec1–6, time1–6	Sectional positions, margins and split times	POST
races.csv, one row per race		
race_id, date, race_no	Race key, meeting date, race number in the meeting	PRE
venue, config, surface	Course, rail configuration, turf or all-weather	PRE
distance, going	Distance in metres, reported track condition	PRE
race_class, horse_ratings, prize	Class, eligible rating band, advertised prize money	PRE
sec_time*, time*	Sectional and cumulative times	POST
_combination, *_dividend*	Winning combinations and tote payouts	POST

end and the numbers reproduced; the code and its output are versioned in the project repository.

A. Data Dictionary

The dataset consists of two relational files published by Daley [21], derived from the public race records of the Hong Kong Jockey Club (HKJC). `races.csv` holds one row per race (6 349 rows, 37 columns) and `runs.csv` one row per horse–race pair (79 447 rows, 37 columns); the two join on `race_id`. The window spans 2 June 1997 to 28 August 2005 and covers 4 405 horses, 186 jockeys and 176 trainers at the Sha Tin and Happy Valley courses. One race in `races.csv` has no corresponding runners and is dropped, leaving 6 348 usable races.

Table III summarises both files. Each field carries an *availability* label: PRE for information published before the race, MKT for market prices, and POST for anything that only exists once the race is decided. That label, not predictive power, determines admissibility. The full dictionary with dtypes, cardinality, ranges and null counts per column is produced automatically by the same pass and kept with the rest of the analysis output in the repository.

B. Data Leakage Audit

Predicting a race from a table that also contains the outcome is trivial and worthless, so the admissibility rule was fixed before any feature was written: *a variable may enter the model only if its value was determined and published before the gates opened for that race*. Applying it column by column across both files discards 48 fields (Table IV), leaving 13 raw pre-race attributes on the runner table and 11 on the race table as the basis for feature engineering.

The target `won` is a deterministic function of `result` ($\texttt{won} = \mathbf{1}[\texttt{result} = 1]$), so keeping `result` would amount to handing the model its label. The sectional blocks describe the race as it unfolds. The payout fields deserve separate mention as the most dangerous leakage here: they sit on the *race* table, so a naive join propagates them silently to every runner, and `win_combination` is literally the saddlecloth number of the winner. They are dropped at load time, before any merge.

`win_odds` and `place_odds` are the *final* tote odds. They are not target leakage in the strict sense, since the price is fixed at post time, but they are excluded from the feature set for three reasons. The closing price aggregates every bettor's information, including late money, so a model built on it largely re-learns the crowd. The resulting system would be useless in deployment, where a prediction is wanted before the market settles. And mixing the market price with a fundamental signal makes our research question unanswerable, since it asks whether pre-race attributes alone carry information. Following the two-stage practice of the racetrack efficiency literature [22], the odds are retained only as an *independent market baseline*.

Most of the signal lives in historical form rather than in the raw row: a horse's prior win rate, jockey and trainer strike rates, days since the last start, class movement. Every such feature uses an expanding window over races strictly earlier in time. The frame is sorted by (entity, `date`) and the aggregate taken as `shift(1).expanding()` within the group, so a race never contributes to its own feature; no statistic is computed over the full table. Debutants make this explicit: 5.5 % of rows have no prior start, so these features are null by construction and receive a "no history" indicator rather than an imputed mean, which would assign an average record to an unraced horse.

The split is temporal *and* grouped by `race_id`. A random split would place runners of the same race on both sides, letting the model exploit the one-winner-per-race constraint, and would let future meetings inform past ones. The cut points are the ones fixed in the experimental protocol of Section IV: meetings up to and including 2003 for training, 2004 for validation and 2005 for test. The partition is checked programmatically to confirm that no race spans more than one split; the count is zero, which holds by construction for any cut placed on a meeting date. Model selection uses forward chaining, never shuffled k-fold. Imputers, encoders and scalers are fitted inside a `scikit-learn Pipeline`

TABLE IV: Excluded fields and the reason for exclusion

Field(s)	Reason
result	The target in ordinal form; won derives from it
lengths_behind, finish_time	Measured at the finish line
position_sec⋆, behind_sec⋆, time⋆	Describe the race while it is run
sec_time⋆ (races)	Sectional times of the race itself
win_combination⋆, place_combination⋆	Encode the identity of the placed runners
win_dividend⋆, place_dividend⋆	Tote payouts, computed from the result
win_odds, place_odds	Not target leakage, but closing market prices; kept only as an external baseline

on training folds only; where high-cardinality identifiers are target-encoded, the encoding is computed out-of-fold with prior smoothing, since a naive target encoding fitted on the whole training set is itself leakage.

C. Data Quality and Imputation

Missingness arises from four distinct mechanisms, and collapsing them into one imputation rule would be a mistake.

Structural absence. The later sectional columns are null for short races, which simply have fewer timing sections: position_sec4, behind_sec4 and time4 are missing in 33 221 rows, rising to 78 151 for the sixth section. These are not missing values, because the quantity does not exist, and imputing them would fabricate a measurement. All are post-race and are dropped regardless.

Systematic absence in a pre-race field. prize is null for 462 races, or 7.3 % of the file, and the gap is not scattered: every affected race falls between 20 July 1997 and 18 March 1998, a contiguous block at the start of the record. This is missingness by collection period, not at random, so mean or median imputation over the whole file would import a late-period price level into early races. Because the split is temporal, the entire block sits inside the training window, which limits but does not remove the problem. prize is therefore kept with an explicit "unrecorded" indicator, and imputation, where needed, is performed within race_class rather than globally, since prize money is largely a function of class.

Sporadic absence. horse_country and horse_type are missing in two rows each and are encoded as an explicit Unknown level. Dropping those rows is specifically rejected: deleting a row removes a runner from a race, changing the size of the field and breaking the one-winner-per-group structure the model depends on. Mode imputation is rejected because it invents a country of origin the model may then use. place_odds is missing in 3 735 rows, in a way related to field size and pool availability rather than at random; the column is excluded anyway, but mean imputation on a variable of that kind would introduce bias rather than remove it.

Sentinels disguised as data. Several columns use 999 to encode "not available", which never appears as a null, so any mean or correlation computed before cleaning is silently corrupted. The affected fields are lengths_behind (89 cells) and the sectional margin and split-time columns (95 further cells). It is worth noting what is *not* a sentinel: race_id and horse_id legitimately take the value 999, and 999 lb is an ordinary declared_weight within the observed 693–1 369 lb range. A naive equality test flags all three, which is why the check is applied only where 999 falls outside the plausible range of the variable.

No univariate imputation is applied to derived historical features, because absence there carries meaning: a debutant is not an average horse. An indicator column is added instead, so a tree can split on it. Median imputation is used only for physical measurements after sentinel removal, and is fitted on training folds only.

D. Outlier Detection

Beyond the sentinels above, three anomalies were found, and only one is an outlier in the statistical sense.

Implausible physical values. Three rows carry a declared_weight below 900 lb, and one of those three is also below 800 lb, which is outside the plausible range for a racing thoroughbred and is treated as a recording error. actual_weight (103–133 lb) and horse_age (2–10 years) fall within legitimate ranges and are kept; the older horses are rare but real.

An impossible category value. One row has draw = 15 although no field in the dataset exceeds 14 runners. With a single occurrence this is a recording error rather than a pattern, but it is reported because a model treating draw as an ordinal would extrapolate from it.

Censoring. win_odds is capped at 99.0, and 9.4 % of runners sit exactly at that ceiling. This is right-censoring, not an outlier: the recorded value is an upper bound, so the ranking among longest-priced runners is arbitrary and the market baseline must say so.

The treatment policy follows from the group structure: rows are never deleted, because removing a runner alters the race it belongs to. Anomalies are neutralised at the cell level, so that the value becomes NaN and a missing indicator is added, and each race keeps its full field.

E. Class Imbalance

Fields range from 4 to 14 runners with a mean of 12.5 and a median of 12, so positives account for 8.0 % of rows, bounded between 7.1 % and 25 % depending on field size. The one-winner-per-race invariant holds in 6 336 races; the remaining 12 (0.19 %) record two winners. These are dead heats, not recording errors. They are retained, since discarding them would bias the sample toward decisive finishes, and the top-1 metric counts a hit when the highest-ranked runner is any of the joint winners.

The imbalance is *structural* rather than a sampling defect: it is a property of the task, not a distortion to correct. That rules out the usual remedies. Synthetic oversampling is rejected on three grounds. A synthetic runner belongs to no race, so it destroys the group structure the evaluation depends on;

it breaks the constraint that win probabilities within a race sum to one; and interpolation is meaningless where `draw`, `jockey_id` and `horse_gear` have no midpoint. Random undersampling fails for the same reason: removing competitors changes how hard the race was to win. Class weighting is admissible for the pointwise variant but distorts the output probabilities, so scores must be recalibrated on the validation split before any log loss is reported. We prefer not to treat the problem as imbalanced at all, but to model the race as a group and normalise scores within it, making the base rate a consequence of field size rather than a nuisance parameter.

The metric plan follows, and it depends on one step that must be stated explicitly: a ranking model outputs scores on no fixed scale, not probabilities. Log loss and Brier score are therefore computed on calibrated outputs, obtained by normalising scores within each race and fitting a calibration map on the validation split. Reporting a raw ranking score as a win probability would be meaningless. With that step in place, accuracy is still the wrong headline metric: a model that never predicts a winner scores 92.0 %. Primary metrics are race-level top-1 accuracy and mean reciprocal rank; calibration is measured by log loss and Brier score; discrimination is reported as ROC-AUC within races. Two reference points anchor the scale. Selecting at random within the field yields 8.1 % top-1 accuracy, and ranking by shortest final odds yields 29.8 %, so the market is 3.7 times better than chance. These two are internal reference points measured on our own data, and they are distinct from the literature baseline of Section III, which is the best result previously published on Hong Kong racing. A model failing to beat 8.1 % is broken; the 29.8 % is the market benchmark, and it is the target the hypothesis of Section I commits us to exceeding.

F. Distributions and Correlations

Field size is concentrated at the upper end, with a median of 12 against a maximum of 14. That governs the base rate above and motivates reporting metrics per race rather than per row. The barrier effect is clear and monotone: runners drawn in stall 1 win 10.5 % of the time against 6.0 % for stall 13, a factor of 1.7 across the range, with the decline steady rather than concentrated at the extremes.

The effect is not uniform across conditions. Splitting by course, the advantage of an inside draw (stalls 1–4) is roughly twice as large at Happy Valley, where those runners win 11.0 % against 7.3 % for the rest of the field, as at Sha Tin, where the same comparison is 8.9 % against 7.2 %; this is consistent with the tighter Happy Valley circuit. By distance, however, the effect is not monotone: the gap is 2.8 points below 1 200 m and 3.1 points above 1 600 m, but only 1.3 points in between. Treating `draw` as a plain ordinal, or interacting it linearly with distance, cannot represent that shape, which is the concrete argument for a draw × distance × venue interaction with distance entered categorically.

Among the pre-race numeric features the notable structure is the near-collinearity between `horse_rating` and `actual_weight`, expected because handicap weight is assigned from the official rating; harmless for gradient-boosted trees, but it must be handled in the logistic-regression baseline by dropping one term or by regularisation.

TABLE V: Diagnosis and resulting decisions

Finding	Action taken
48 columns encode post-race information	Dropped at load time, before any join
Payout fields sit on the race table	Removed before the `race_id` merge
Closing odds beat chance by 3.7×	Excluded as a feature; kept as external baseline
Historical form is the main signal; 5.5 % debutants	Expanding windows with `shift(1)`; explicit no-history indicator
Runners of one race are not independent	Temporal split grouped by `race_id`; zero races span two splits
`999` used as a sentinel in 184 cells	Converted to `NaN`, but only where 999 is outside the variable's plausible range
`prize` missing for a contiguous 1997–98 block	Indicator column; imputation within `race_class`, never global
3 implausible weights, 1 impossible `draw`	Cell-level neutralisation; rows never deleted
`win_odds` censored at 99.0 (9.4 % of runners)	Documented as a limit of the market baseline
12 dead heats, 1 race without runners	Dead heats retained with an adjusted top-1 rule; empty race dropped
8.0 % positives, one winner per race	No resampling; within-race normalisation
`horse_rating` collinear with `actual_weight`	Regularisation in the linear baseline

G. Findings and Derived Actions

Table V condenses the analysis into the concrete decisions it produced, each of which is carried into the experimental protocol of Section IV.

VI. Ethical Considerations

Two features of this problem make an early ethical review necessary rather than ceremonial. The application domain is a regulated gambling market, and the underlying activity has contested animal-welfare implications. Both risks are properties of the domain and cannot be engineered away; what can be decided in advance is the scope of the artefact and the claims made about it. This section states the risks we identified before training and the specific limits adopted in response. The AI usage disclosure covers the separate question of tooling, and the Stage 3 report revisits these issues with the deployed agent in view.

A. Scope of the artefact

The system estimates a win probability for each runner from information published before a race, and ranks the field accordingly. It is a retrospective study on historical records from 1997–2005. It does not consume live odds, holds no connection to any betting operator, and includes no staking, bankroll or bet-selection component. The objective function is probabilistic accuracy, not return on investment: the model is selected and reported on ranking quality together with log loss and Brier score computed on calibrated outputs, and never on simulated profit. This is a methodological choice as much as an ethical one, since optimising for return would make results

dependent on assumptions about the tote pool that the dataset does not support. It is also the boundary that keeps the project an exercise in probability estimation rather than a wagering tool.

B. Gambling-related harm

The most obvious risk is dual use. A calibrated win probability is directly actionable in a betting market, and Hong Kong hosts one of the world's largest per-capita betting turnovers, with problem gambling recognised as a public-health concern there. Publishing an accurate model lowers the cost of participating in that market for anyone who finds it.

We do not claim this risk is eliminated; the mitigations bound it. The repository carries an explicit non-deployment statement and no interface to any betting platform. No trained model weights are published in a form intended for reuse as a tipping service. The Stage 3 agent is scoped to explaining and comparing predictions, describing which pre-race attributes moved a given estimate and how the model's ranking compares with the market's. It declines requests for staking advice, stake sizing or bet recommendations, with that refusal treated as a functional requirement to be tested rather than a disclaimer in a README. Where results are compared against market prices, the comparison is framed as a test of informational efficiency, following the standard treatment in that literature, and not as evidence that a profitable strategy exists.

A residual risk remains and we state it plainly: any work that improves the informational efficiency of a betting market is of some use to people betting in it. Declaring the limit does not remove it.

C. Animal welfare

Thoroughbred racing carries documented welfare concerns, including racing-related injury and the fate of horses after their careers end. This project does not model, optimise or recommend anything at the level of an individual animal: it makes no recommendation about how a horse should be trained, campaigned or run, and it produces no incentive directed at a trainer or owner. It reuses records of races that were run two decades ago. That said, the work belongs to an ecosystem whose economics rest on the racing industry, and we note the tension rather than resolve it. Nothing in the analysis should be read as a position on whether the sport ought to exist.

D. Identifiable individuals

Jockeys and trainers appear as pseudonymous integer identifiers, but the pseudonymisation is weak: because the underlying race records are public, an identifier can be re-linked to a named professional by matching dates, venues and results. The features we build include jockey and trainer strike rates, so the model assigns what amounts to a performance score to real people whose livelihood depends on their reputation.

Three limits follow. Identifiers stay pseudonymous in every published output, including figures and appendices. No ranking of individual jockeys or trainers is produced or published, even though the intermediate features would permit one. Feature-importance analysis is reported at the level of feature families, such as "jockey history", "barrier" and "weight", rather than for specific individuals. Horse identity raises no comparable concern, and `horse_id` is used without restriction.

E. Representativeness and the limits of the claim

The data come from two courses under a single regulator, within a fixed historical window, and under Hong Kong's particular handicap and class system. Results generalise to that setting and no further; presenting them as a general statement about racing would be an overclaim. There is also drift within the window itself, including rule changes, track work and turnover among jockeys and trainers. That is one reason the evaluation uses a temporal split (Sec. V-B): it makes degradation over time visible instead of averaging it away, at the cost of a harder test set. Any bias present in the historical records, including systematic differences in how horses of different origins or classes were rated and handicapped, is reproduced by a model fitted to them.

F. Calibration and overclaiming

A win probability presented without its uncertainty invites misuse, and the temptation in this domain is to read a poorly calibrated score as a statement of fact. Racing outcomes carry substantial irreducible uncertainty: the best achievable performance is far from perfect prediction, and a model reporting high confidence is more likely miscalibrated than skilled. We therefore report calibration explicitly through log loss, Brier score and reliability diagrams alongside ranking metrics, and state the irreducible-uncertainty limit in the abstract and the conclusions rather than only in an appendix.

G. Provenance and licensing

The dataset is a public Kaggle release derived from race records published by the HKJC. It contains no personal data of bettors and no data subject to consent requirements. It is released under a CC0 1.0 Universal public domain dedication, which places no restriction on reuse and requires no attribution; we nonetheless cite the source [21], since scholarly attribution is an obligation separate from the licence.

AI Usage Disclosure

Generative artificial intelligence tools were used as supporting tools during the development of this work. Their use was primarily focused on exploratory analysis, literature review, methodological discussion, and editorial support. In the literature review, AI tools were used to locate relevant information within papers, identify reported metrics and results, compare methodological approaches across studies, identify limitations explicitly stated by the authors, and suggest potential connections between related works. AI assistance was also used to organize bibliographic information and support the preparation of citations and references.

AI tools were additionally used to support the preliminary examination of the project datasets, including the identification of relevant variables, possible data-quality issues, and considerations regarding feature availability and data preprocessing. In the development of the research approach, AI was used to discuss alternative machine-learning formulations, evaluation strategies, and methodological considerations relevant to the prediction problem. Where appropriate, AI assistance was also used to propose or review small scripts for verification tasks, such as checking dataset properties or document consistency, and to identify possible inconsistencies between different sections of the project documentation.

The use of AI did not replace the authors' analysis or academic judgment. AI- generated interpretations, methodological suggestions, bibliographic information, and technical observations were treated as preliminary assistance rather than authoritative evidence. Similarly, references identified indirectly through another publication were treated as secondary citations and independently reviewed its accuracy and potential value to our reasearch.

All relevant AI-assisted content was subsequently reviewed by the authors. Bibliographic references and reported results were checked against their corresponding sources, dataset-related observations were manually verified, and methodological suggestions were evaluated according to the objectives and constraints of the project. The authors retain full responsibility for the accuracy of the final document, the correctness of its citations, the interpretation of the literature, the methodological decisions, the experimental results, and the conclusions presented in this work.

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